SimCRC Chile — Calibration Results

Adenoma Model (Female) · v0.13.0 · April 6, 2026

SimCRC team

Calibration Overview

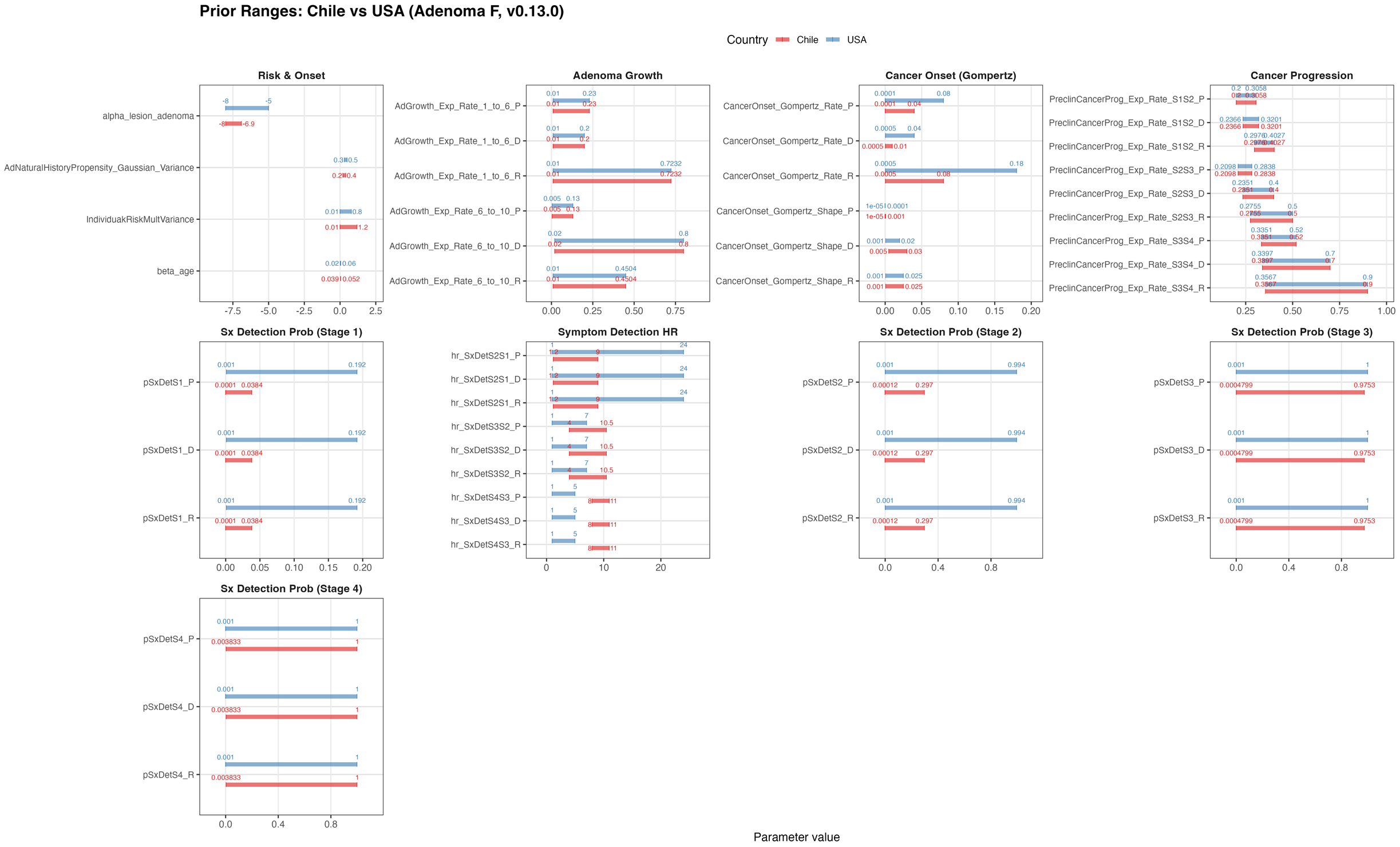
Model: SimCRC — Adenoma (Female), Chilean population

Version: v0.13.0 · Calibrated April 6, 2026

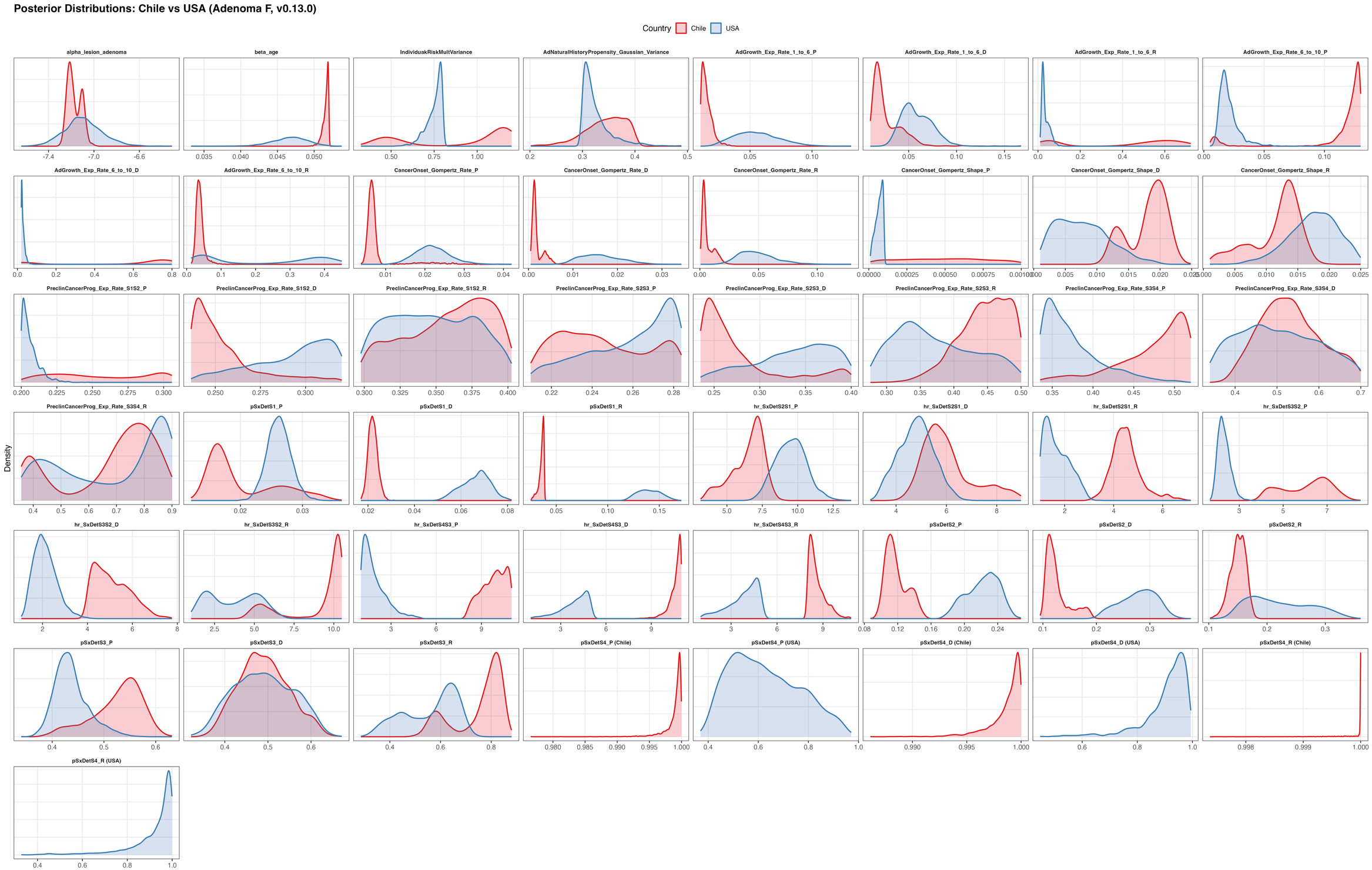
Method: BayCANN (Bayesian Calibration with Artificial Neural Networks)

- Priors derived from the **US model**, adjusted per Nicolas' suggestions
- Modified parameters: `alpha_lesion_adenoma`, hazard rates, preclinical-to-clinical detection probabilities
- Calibration targets from **Chilean epidemiological data**
- **4 MCMC chains** used for posterior sampling

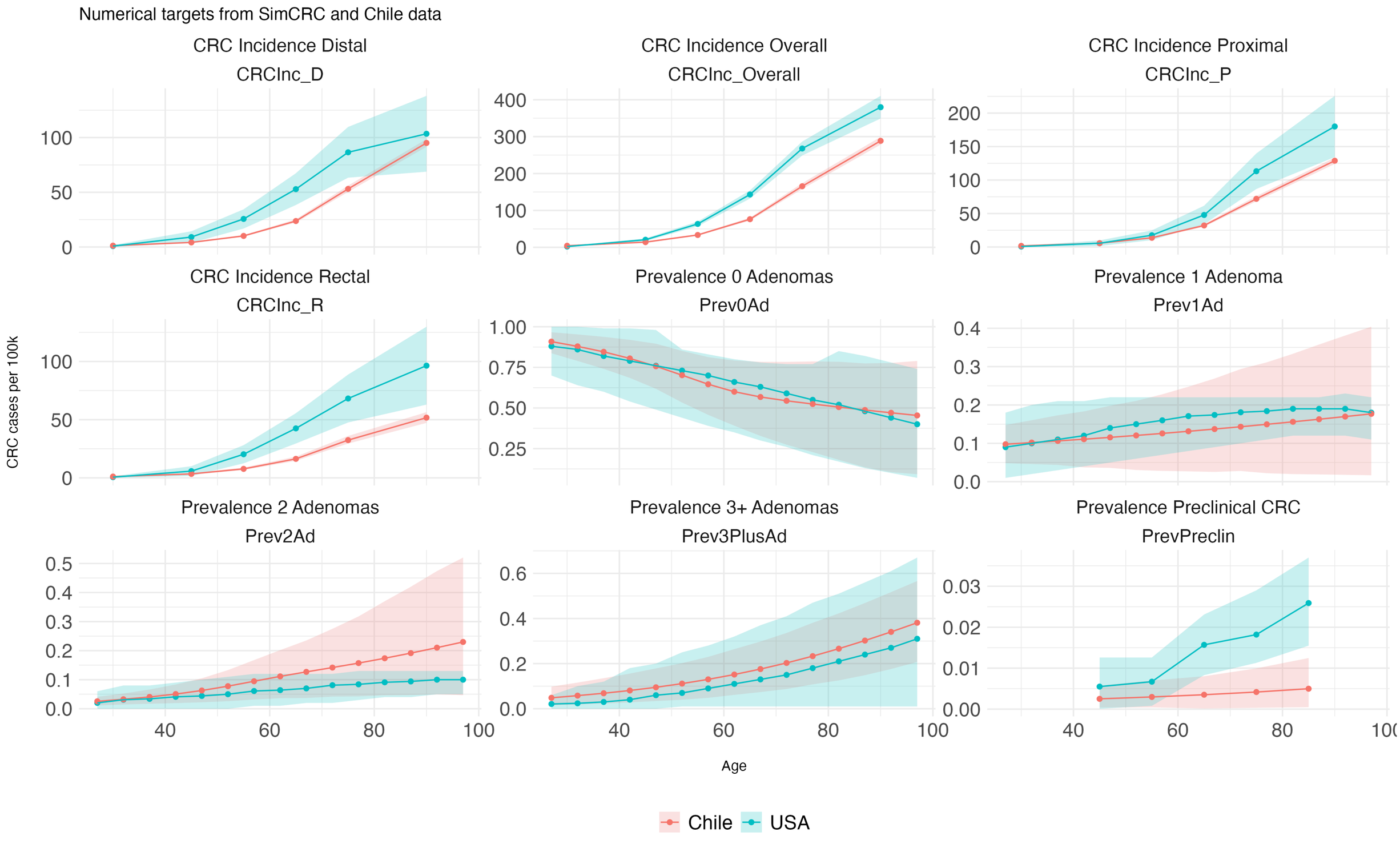
Prior Ranges — Chile vs USA



Posterior Distributions — Chile vs USA

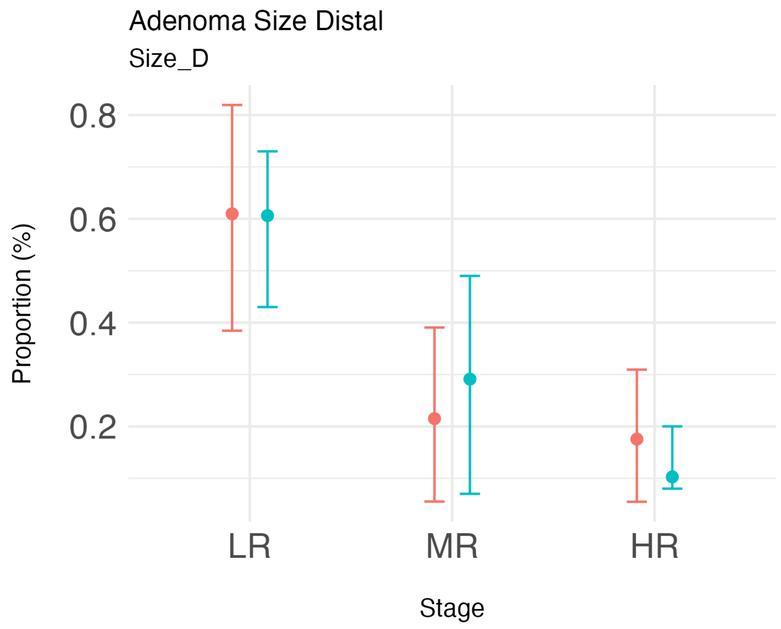


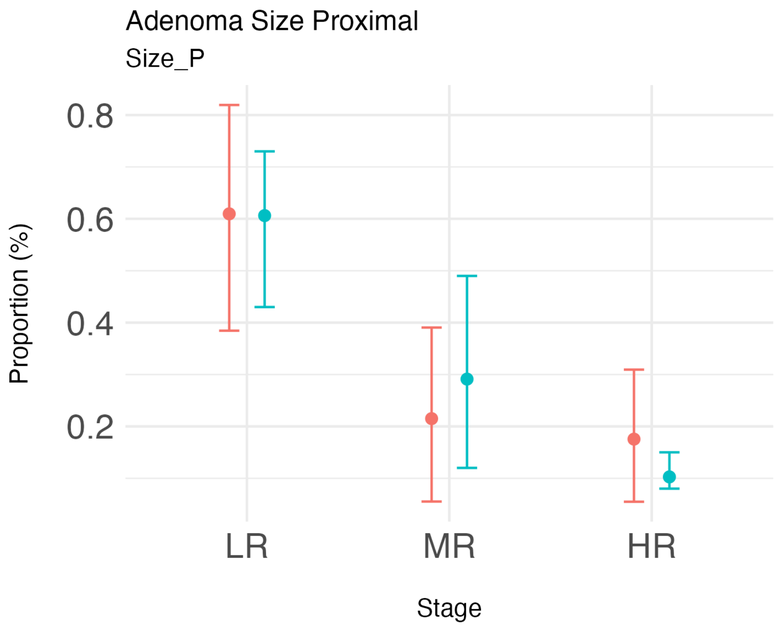
Calibration Targets — Numerical (Chile vs USA)

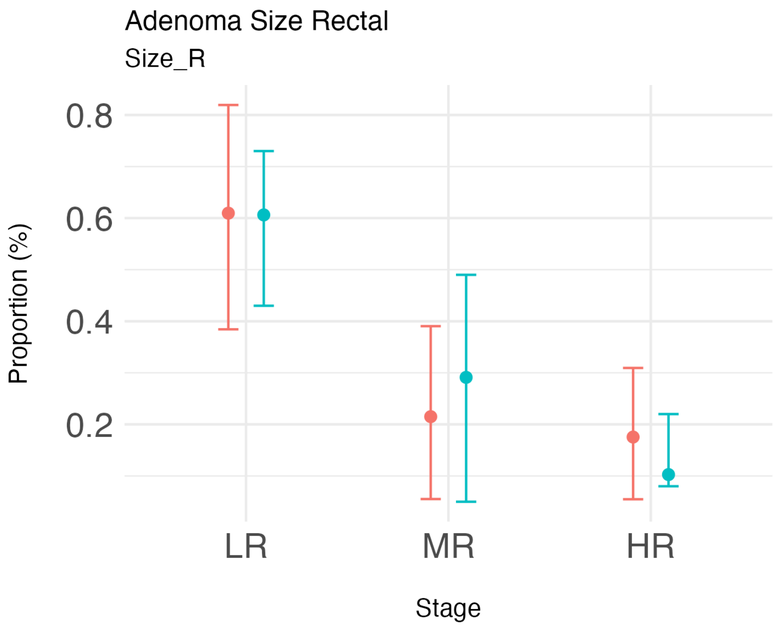


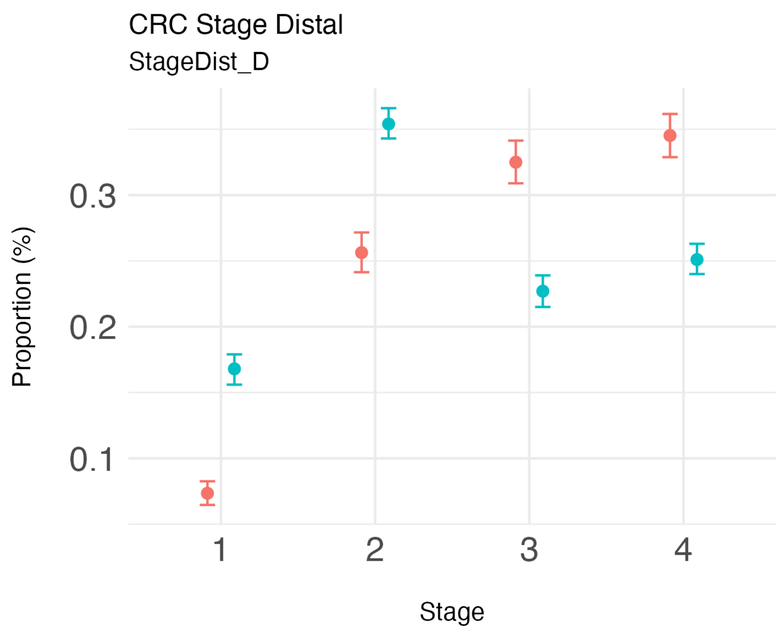
Calibration Targets — Categorical (Chile vs USA)

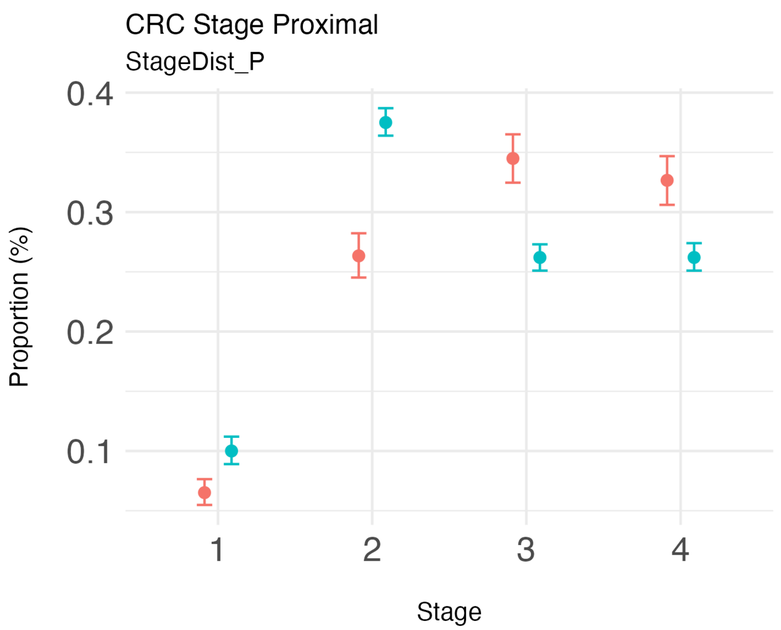
Categorical targets from SimCRC and Chile data

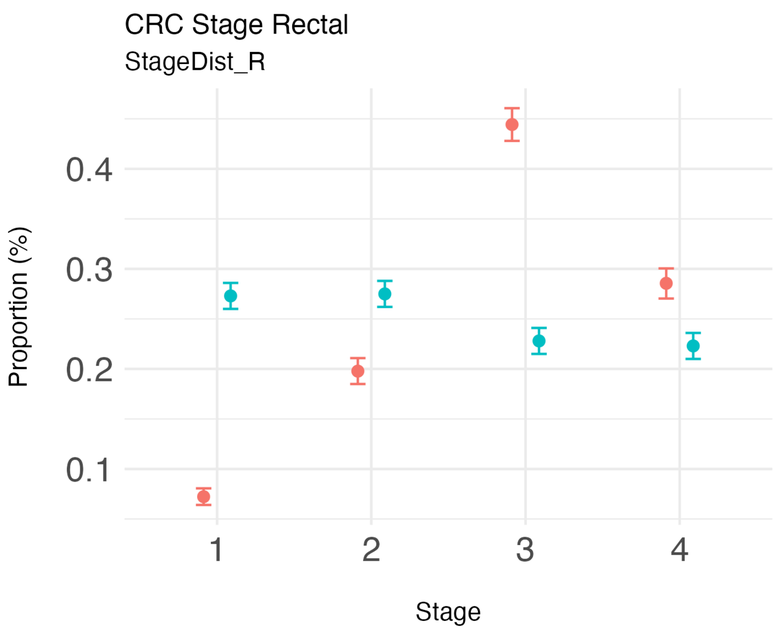








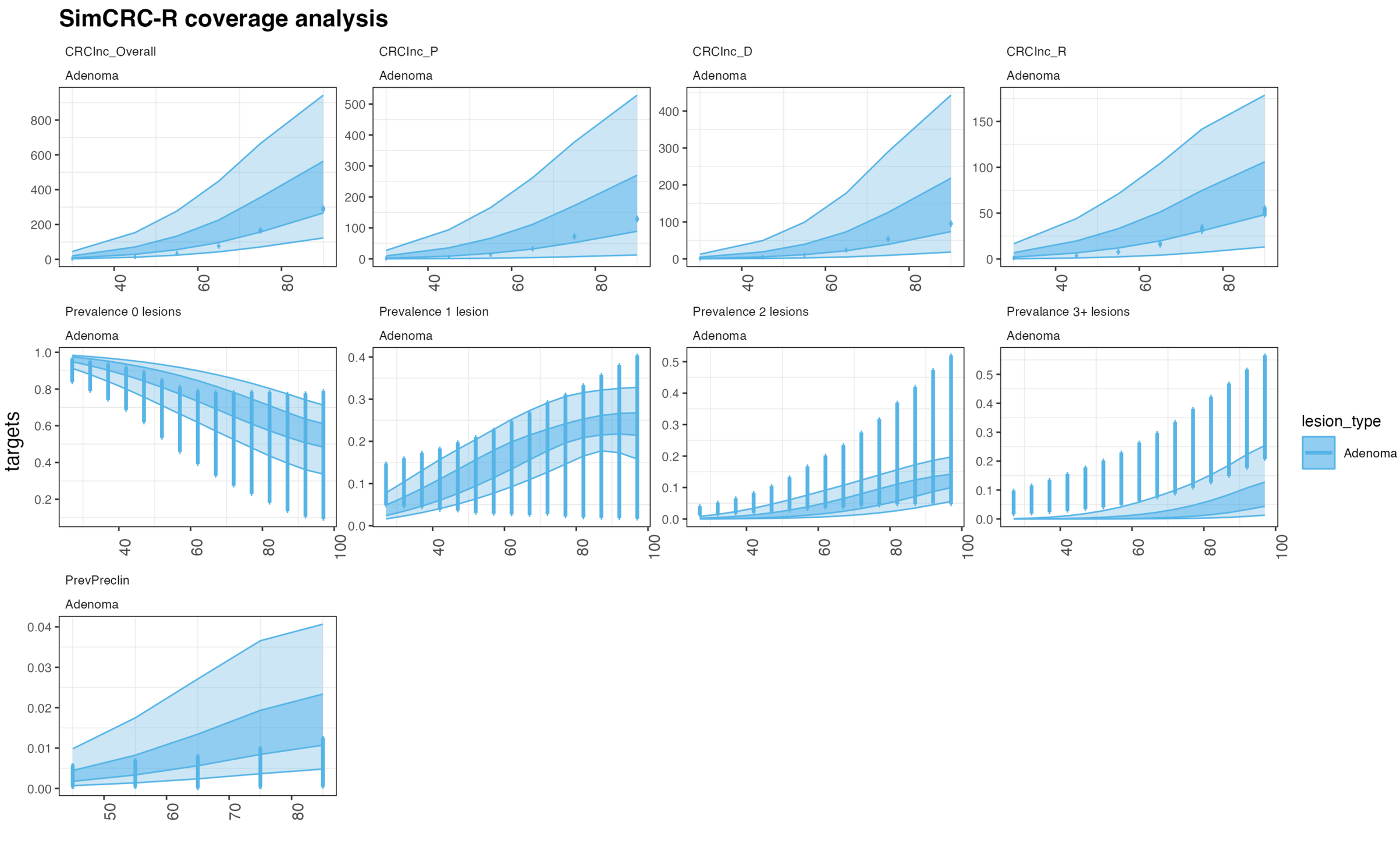




Chile USA

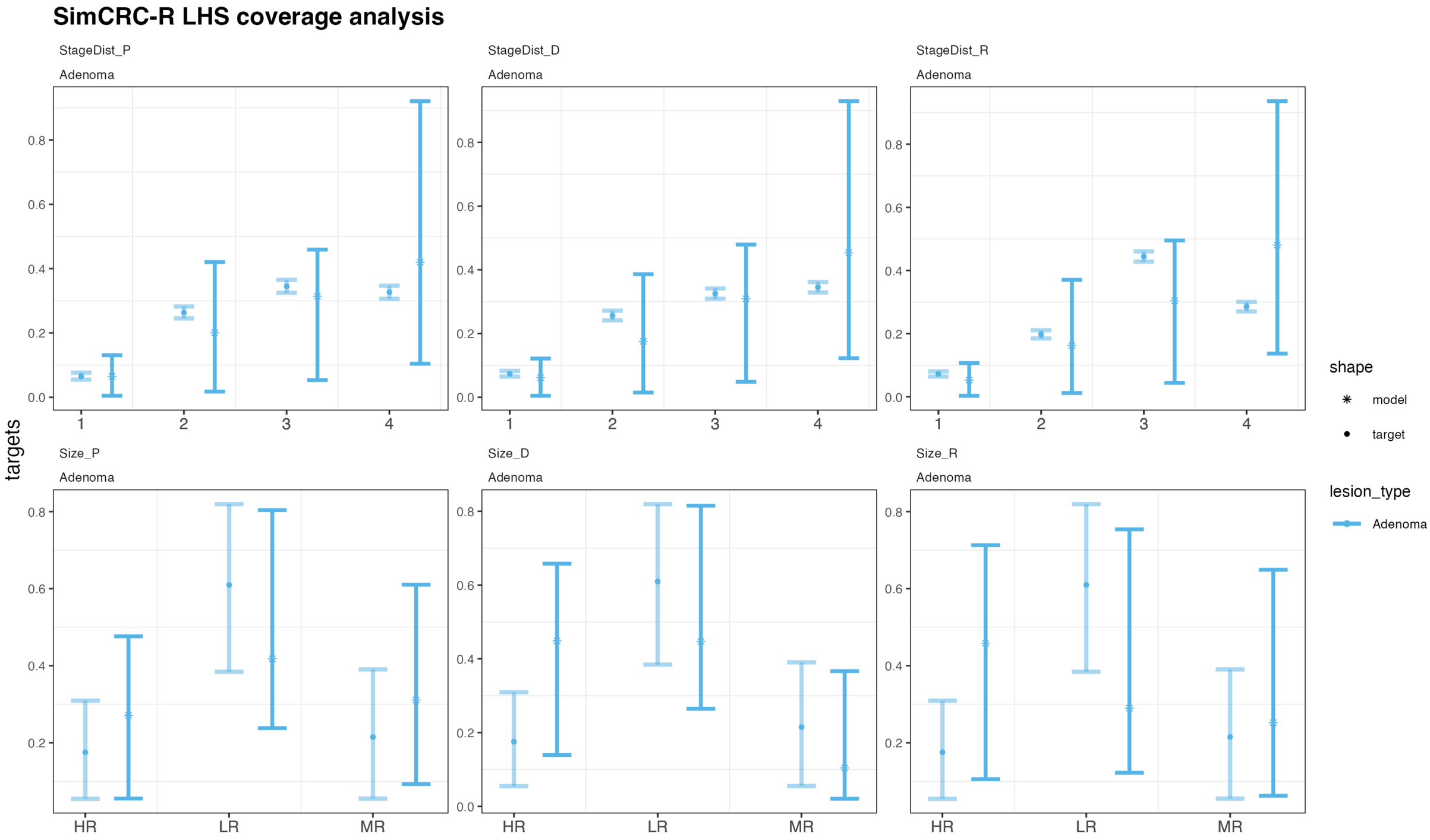
Coverage Analysis — Numerical Targets

LHS coverage for CRC incidence (overall, proximal, distal, rectal), adenoma prevalence (0–3+ lesions), and preclinical prevalence.



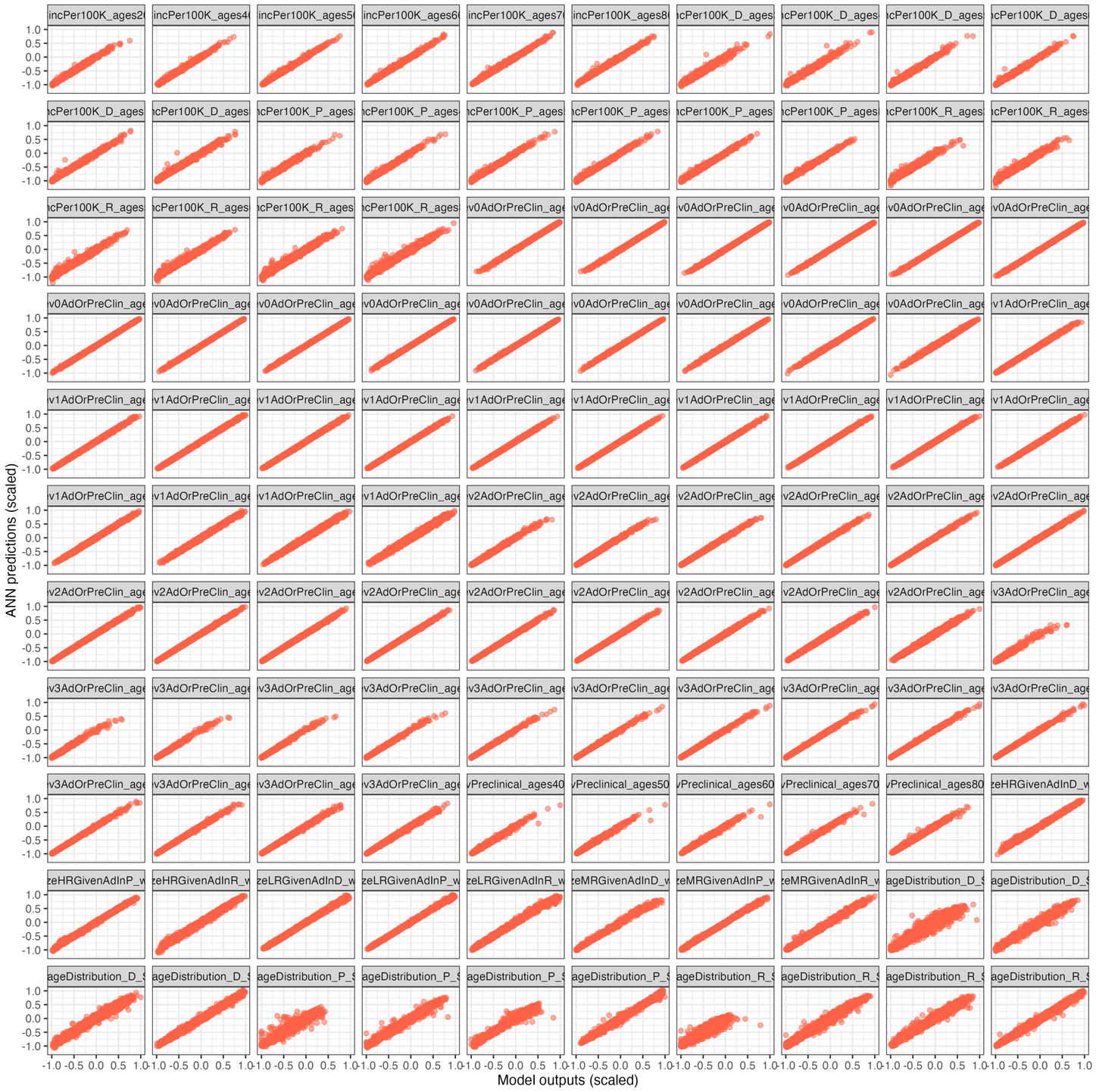
Coverage Analysis — Categorical Targets

LHS coverage for stage distribution and size distribution by location (proximal, distal, rectal).



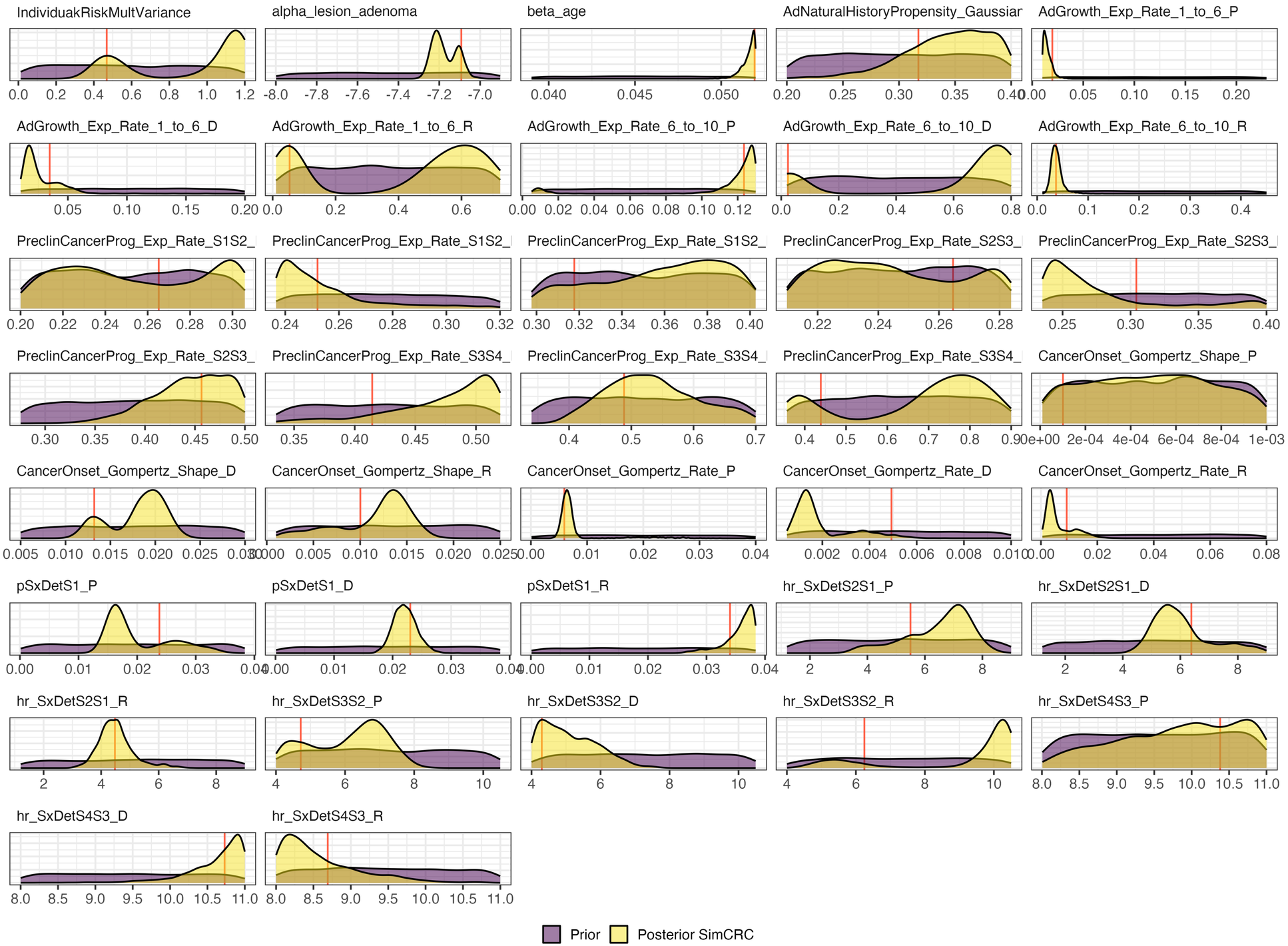
ANN Prediction Performance

Neural network predicted vs. actual model outputs across all calibration targets.



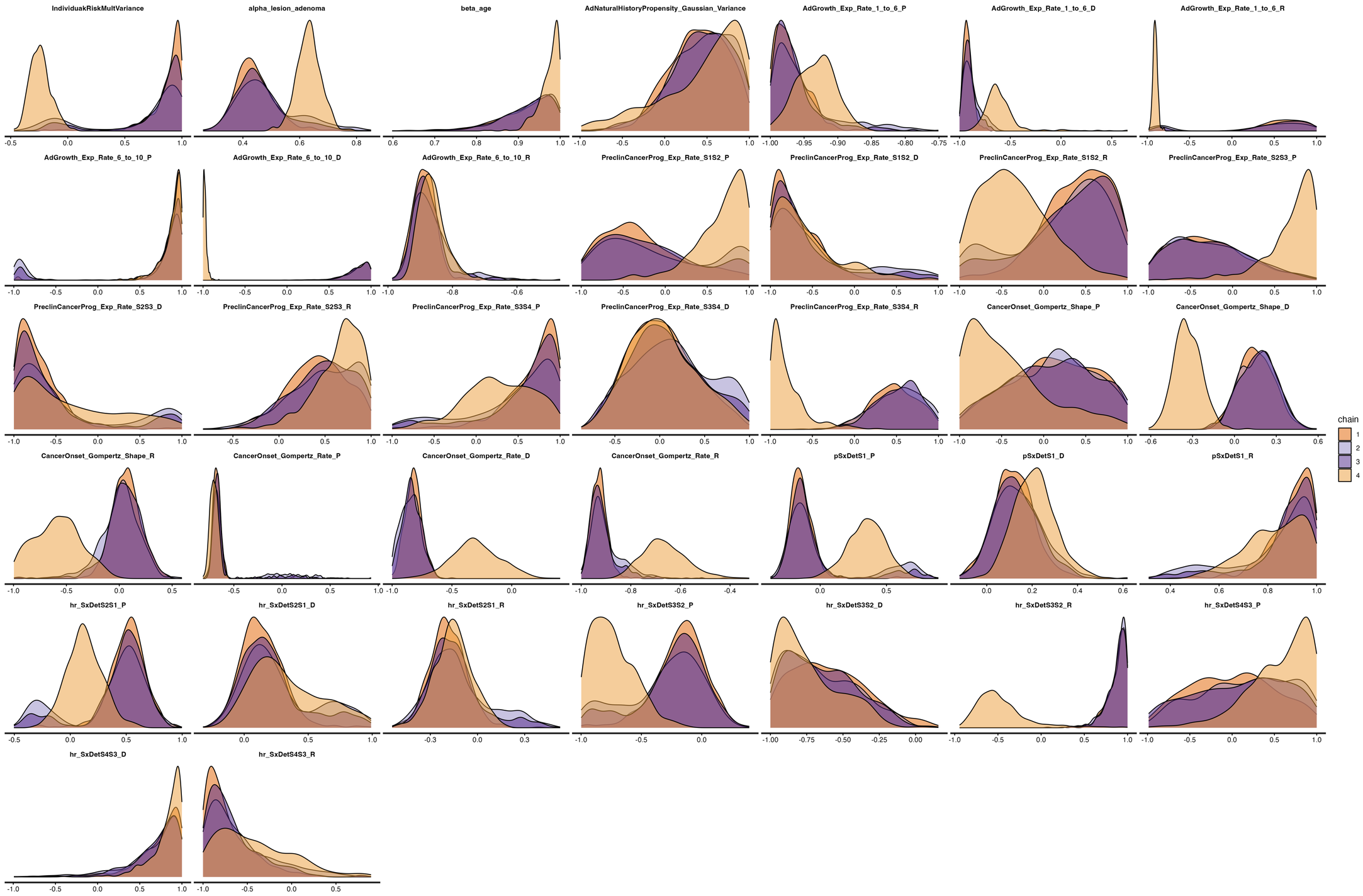
Prior vs Posterior Distributions

Comparison of prior (purple) and posterior (gold) distributions for all calibrated parameters.



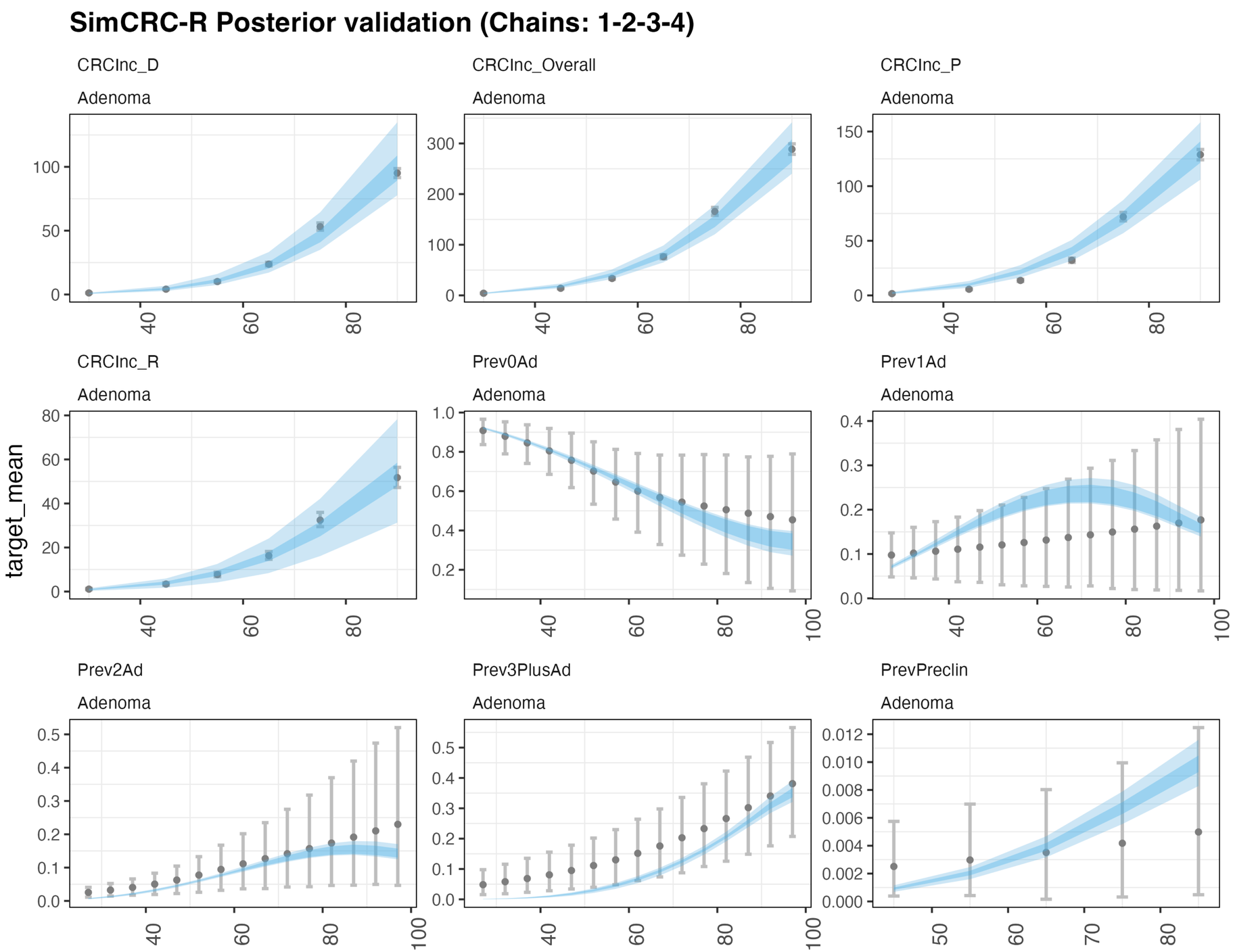
Posterior Distribution by Chain

Chain-level posterior densities — checking convergence across the 4 MCMC chains.



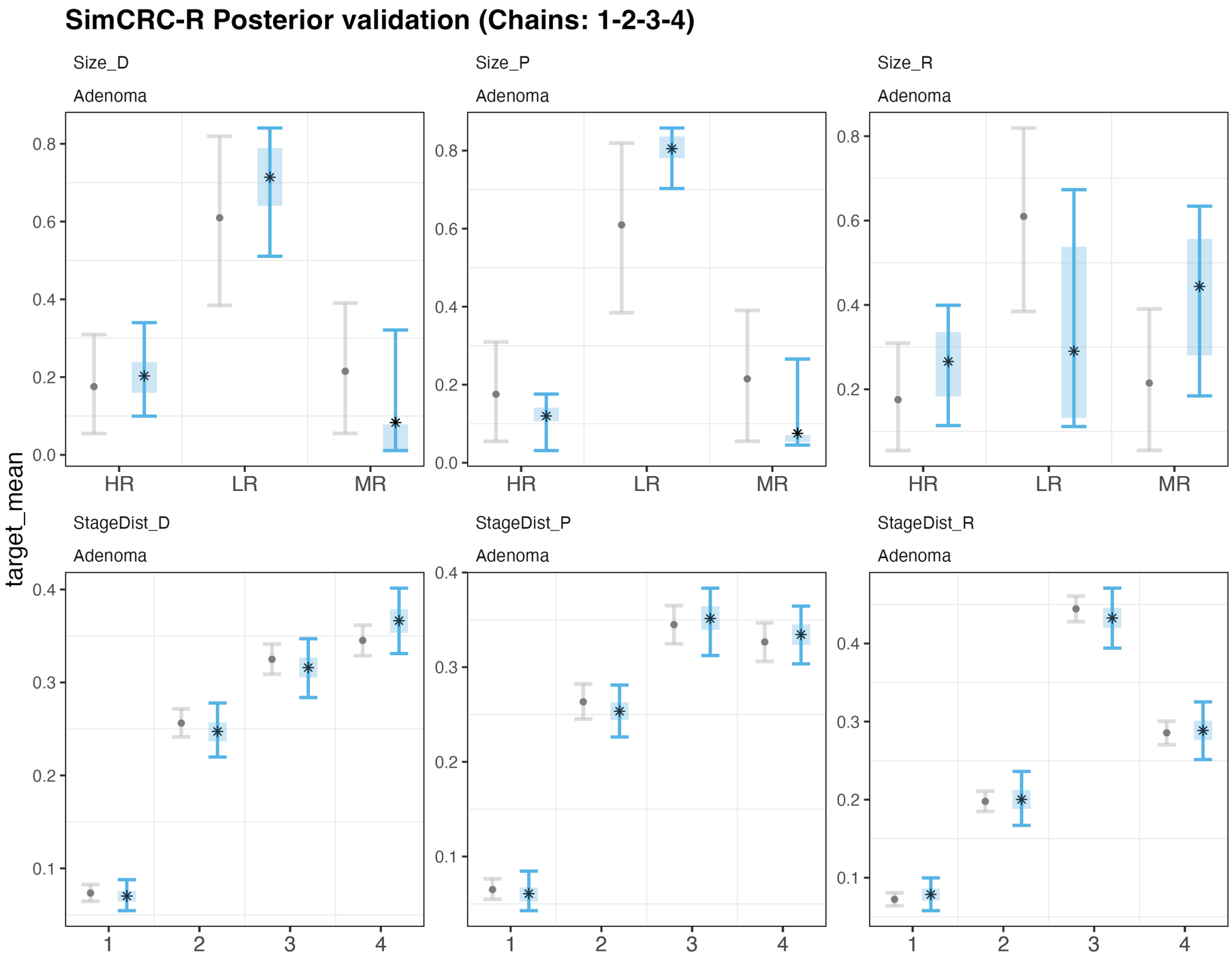
Internal Validation — Numerical Targets

Posterior validation for CRC incidence and adenoma prevalence targets across age.

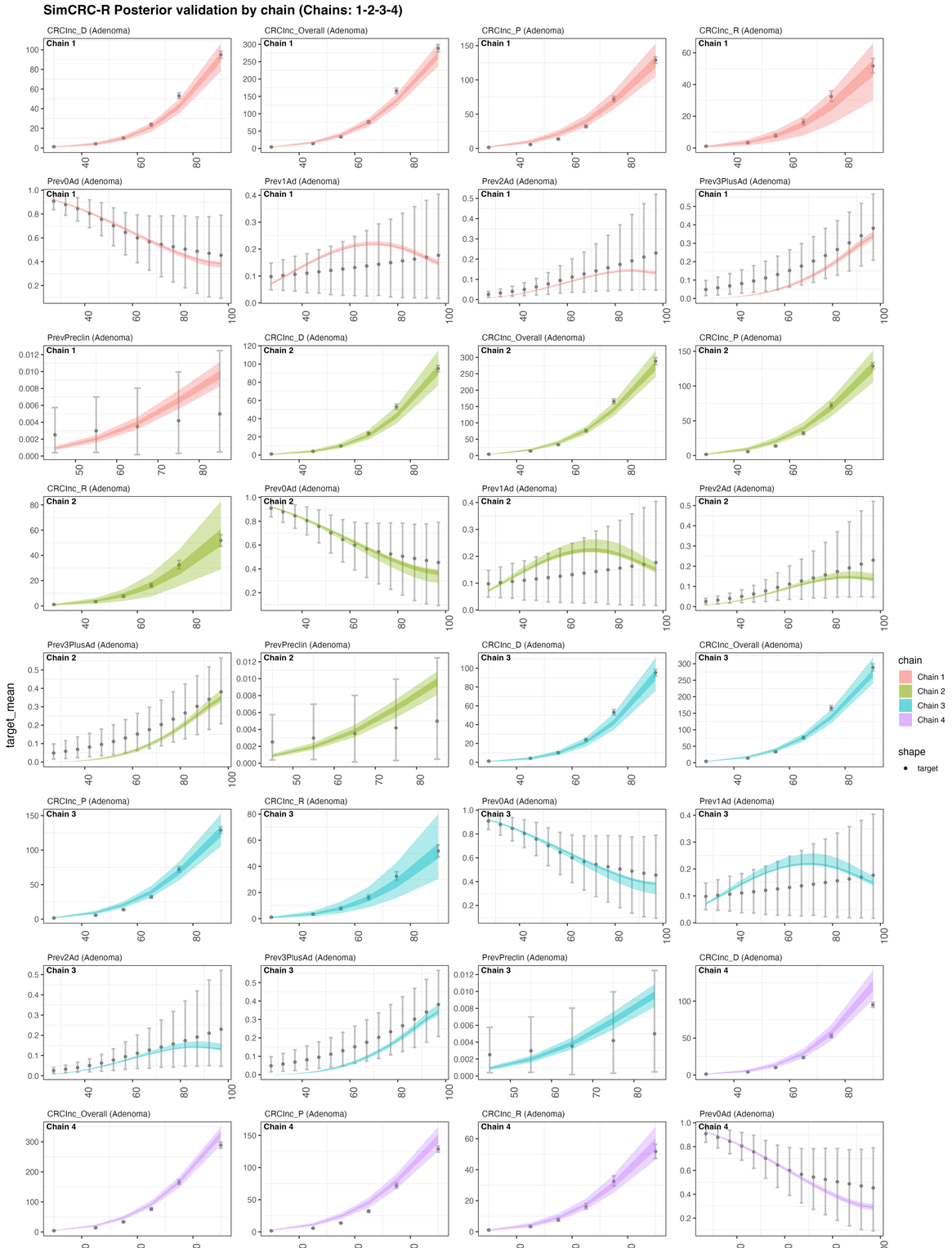


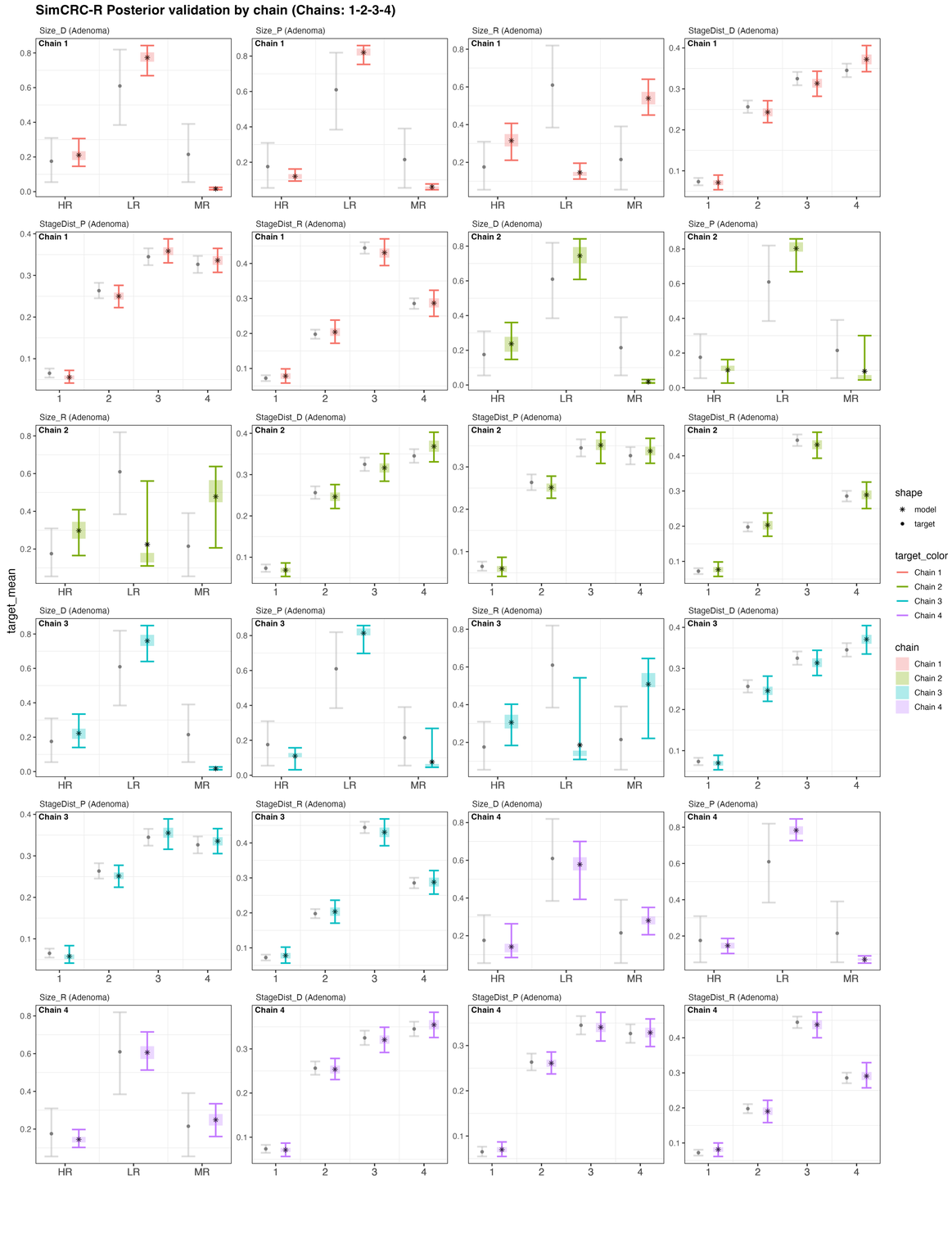
Internal Validation — Categorical Targets

Posterior validation for size distribution and stage distribution targets.



Internal Validation by Chain



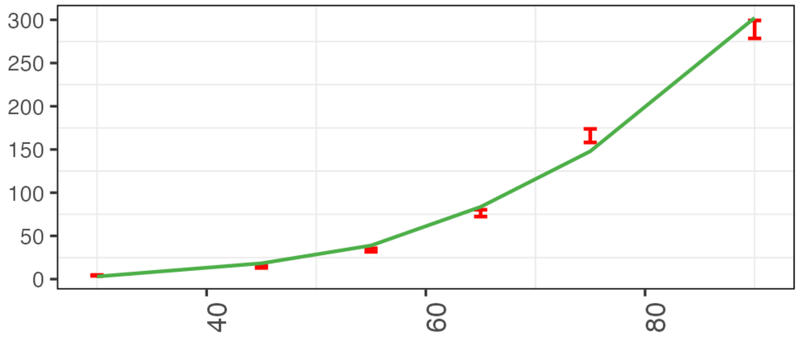


Calibrated Parameter Set — Numerical (Min Absolute Error)

SimCRC-R calibrated parameter set validation (Set: Min_AbsolutErr)

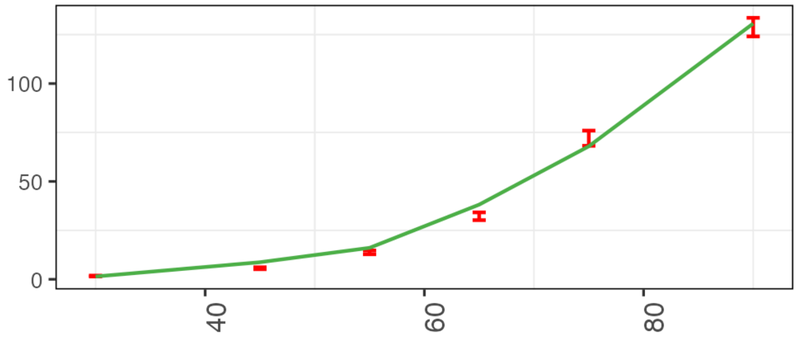
CRCInc_Overall

Adenoma



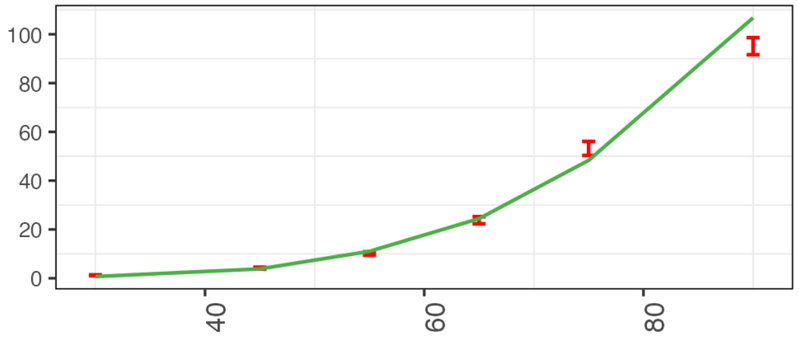
CRCInc_P

Adenoma



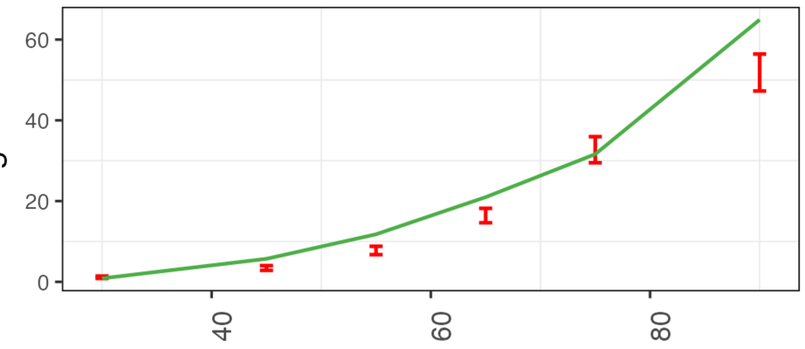
CRCInc_D

Adenoma



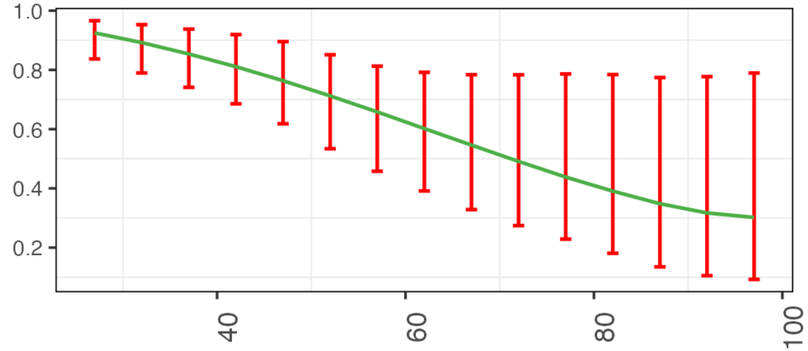
CRCInc_R

Adenoma



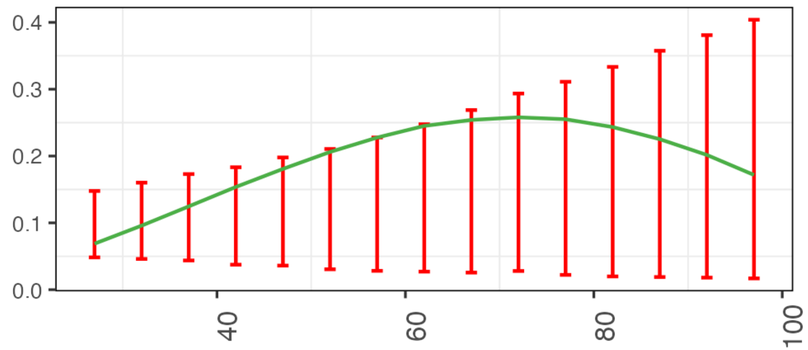
Prevalence 0 lesions

Adenoma



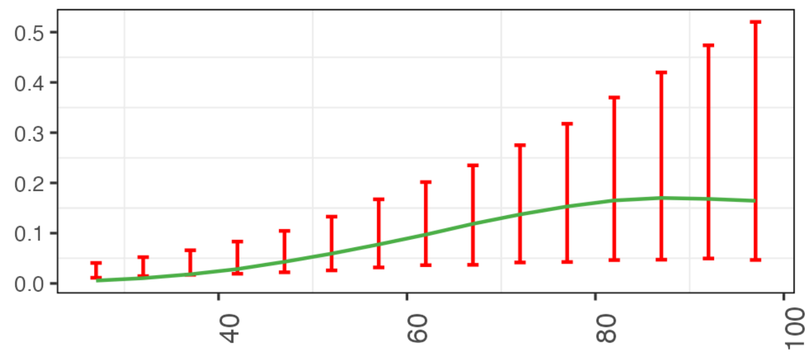
Prevalence 1 lesion

Adenoma



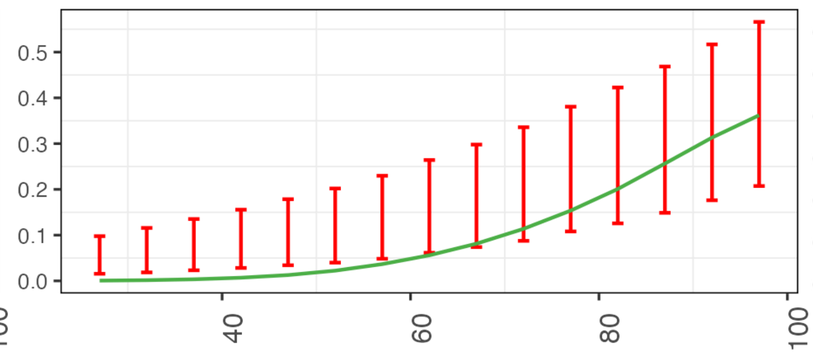
Prevalence 2 lesions

Adenoma



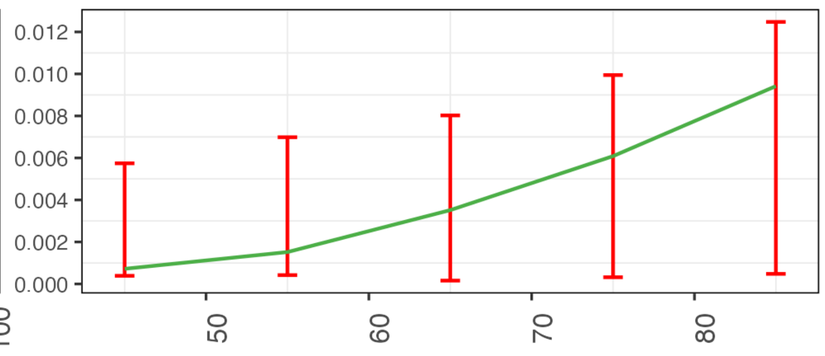
Prevalance 3+ lesions

Adenoma

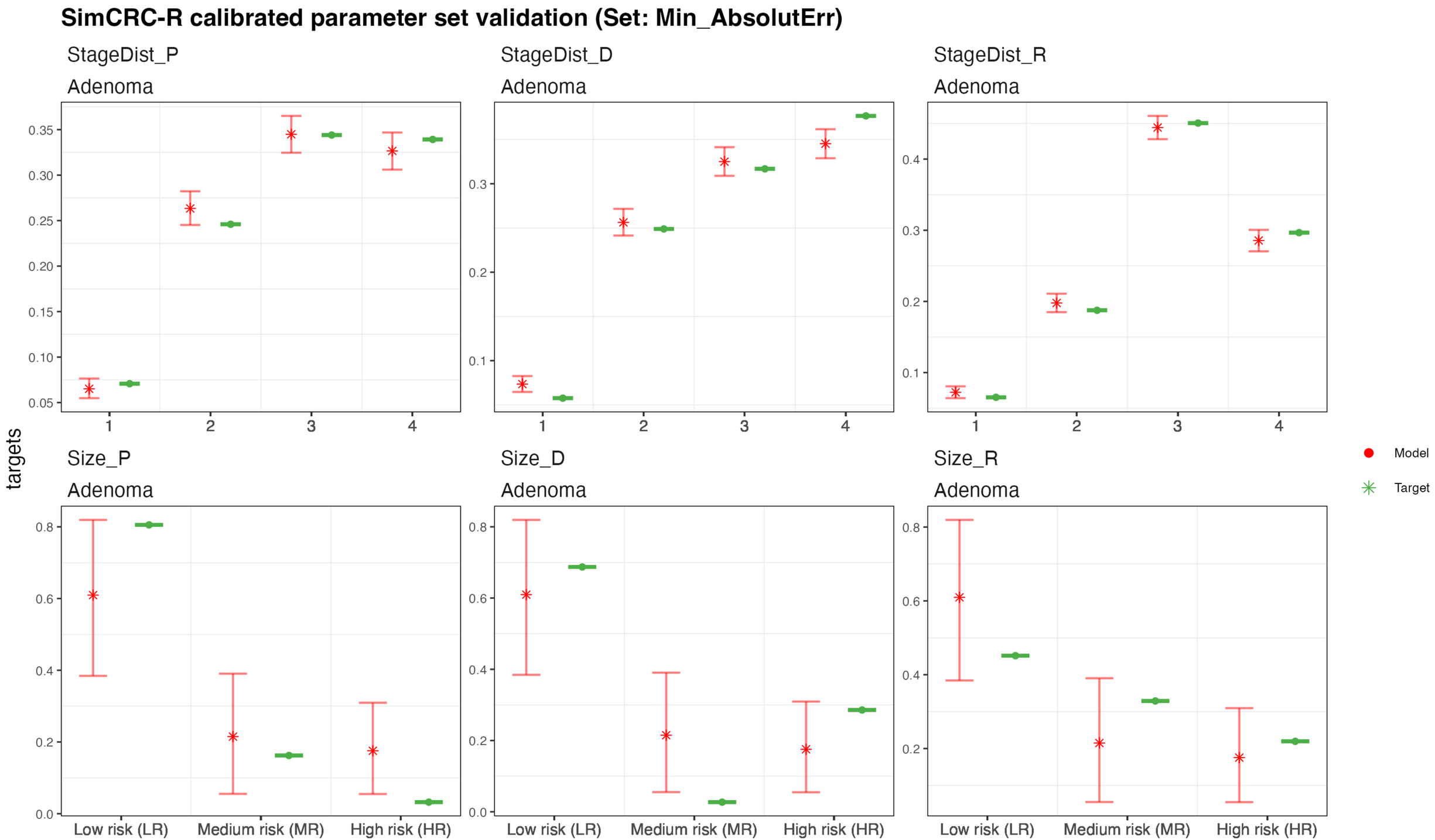


PrevPreclin

Adenoma

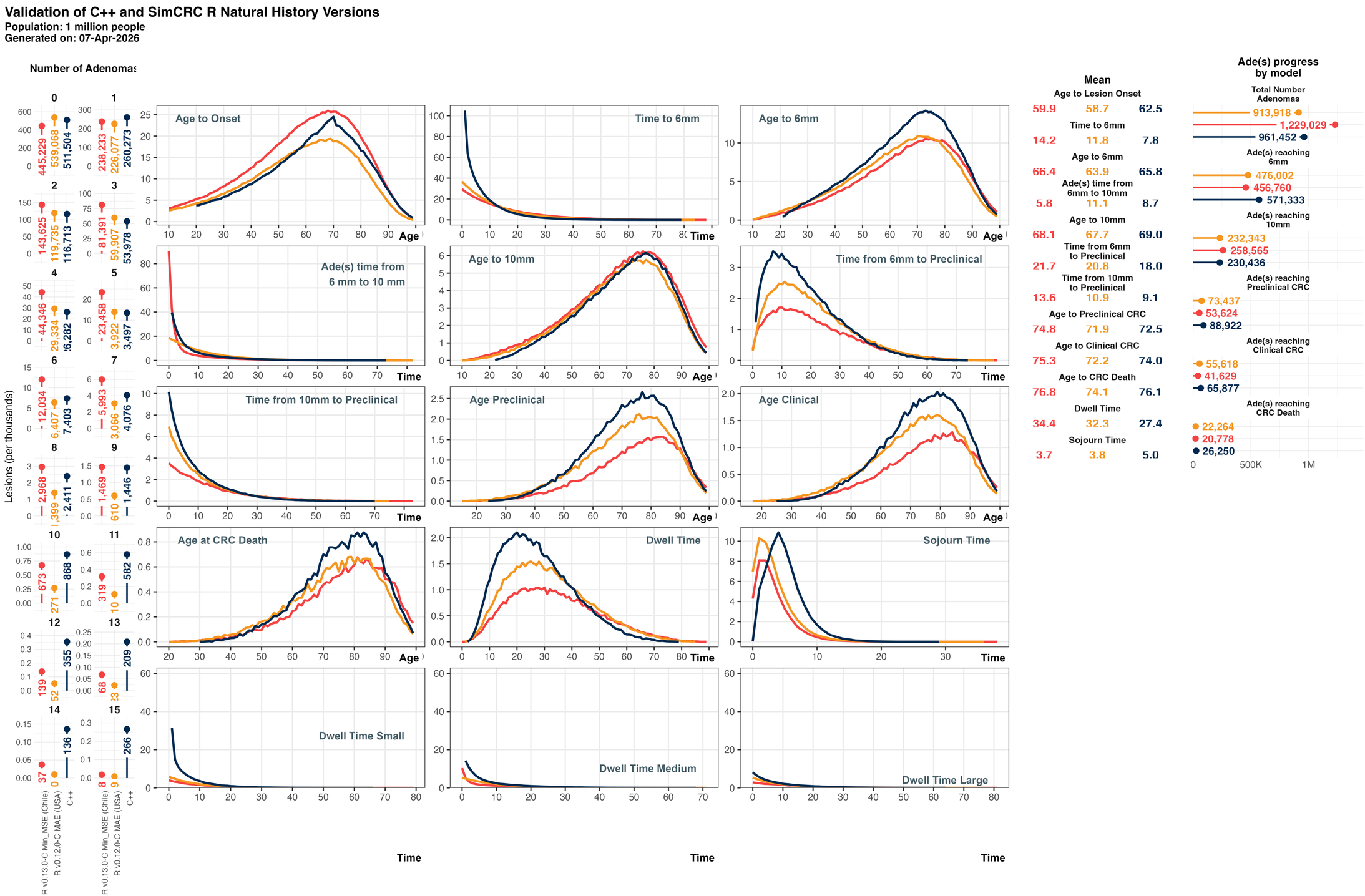


Calibrated Parameter Set — Categorical (Min Absolute Error)



C++ and SimCRC-R Validation

Validation of C++ and SimCRC-R natural history versions — Chile (v0.13.0) vs USA (v0.12.0), minimum MAE parameter set (Female, 3rd).



Screening Cost Update — v2 vs v3 (CLP)

Updated to 2026 Chilean costs (`screen_costs_v3.csv`). Key changes:

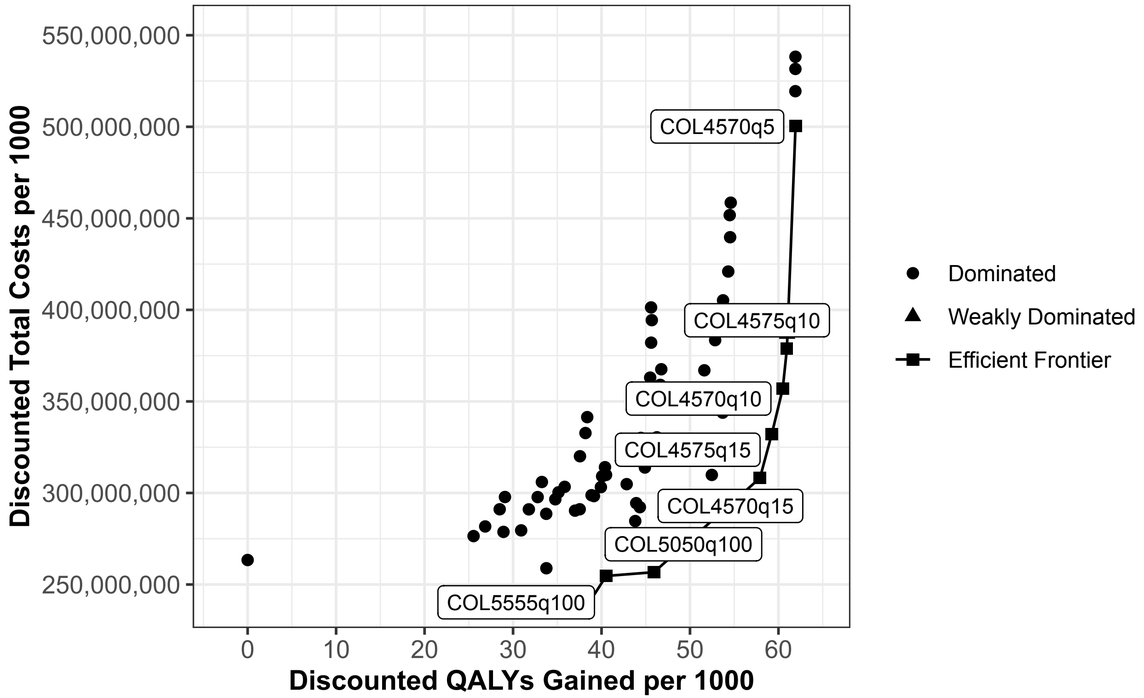
Procedure	v2	v3	Change
FIT	14,220	21,510	+51%
SIG	77,100	118,064	+53%
Screening colonoscopy (no polypectomy)	121,664	181,271	+49%
Diagnostic colonoscopy (no polypectomy)	105,885	157,762	+49%
Surveillance colonoscopy (no polypectomy)	119,993	178,781	+49%
Colonoscopy with polypectomy	144,028	426,078	+196%
Fatal perforation	1,134,170	3,355,203	+196%
Serious GI complication	1,134,170	3,355,203	+196%
Other GI complication	808,045	2,390,433	+196%
Cardiovascular complication	885,641	2,619,984	+196%

Cost-Effectiveness — All Strategies (v2 vs v3)

Costs v2

All Strategies (v2)
WTP = $16M per QALY

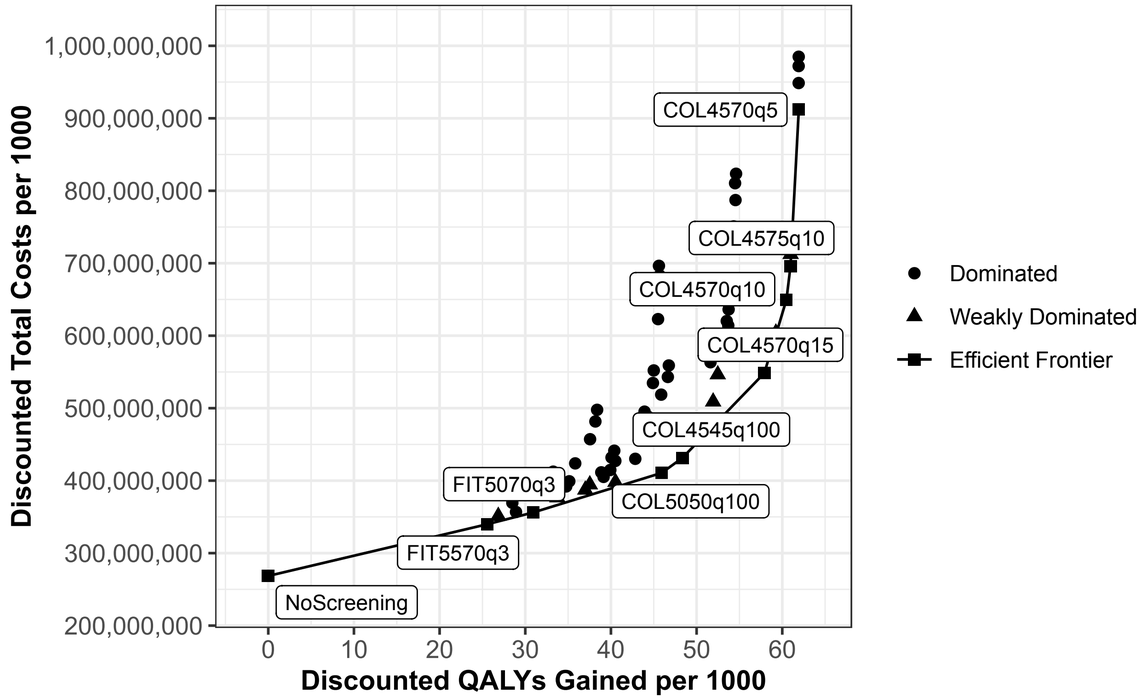
	Cost (M$)	QALYs	ICER (M$)
COL5555q100	254.7	40.5	—
COL5050q100	256.7	45.9	0.38
NoScreening	263.4	0.0	—
COL4545q100	267.9	48.3	—
FIT5570q3	276.5	25.5	—
FIT5070q3	279.6	30.9	—
COL4570q15	308.1	57.9	4.29
COL4575q15	332.2	59.3	17.92
COL4570q10	357.0	60.5	20.07
COL4575q10	378.9	61.0	46.22
COL4570q5	500.4	61.9	125.14



Costs v3

All Strategies (v3)
WTP = $16M per QALY

	Cost (M$)	QALYs	ICER (M$)
NoScreening	268.5	0.0	—
FIT5570q3	339.8	25.5	2.79
FIT5070q3	356.1	30.9	3.02
COL5555q100	398.0	40.5	—
COL5050q100	410.7	45.9	3.64
COL4545q100	431.5	48.3	8.61
COL4570q15	548.5	57.9	12.24
COL4575q15	604.1	59.3	—
COL4570q10	649.6	60.5	39.23
COL4575q10	695.8	61.0	97.60
COL4570q5	912.0	61.9	222.57

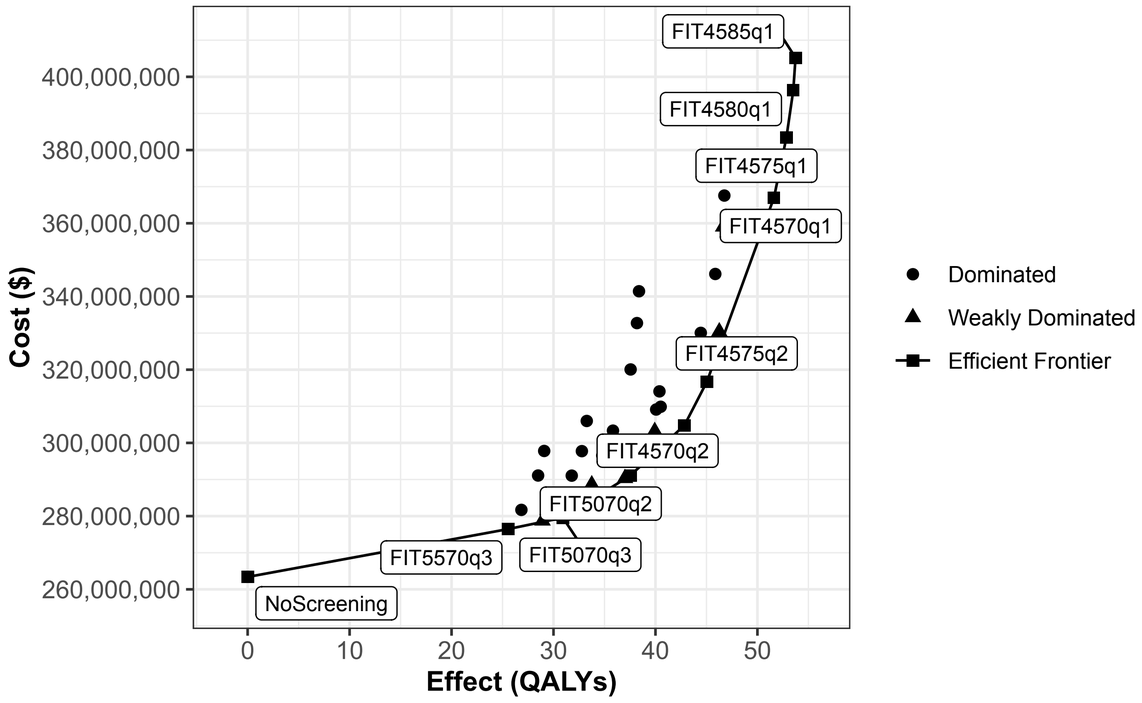


Cost-Effectiveness — FIT Strategies (v2 vs v3)

Costs v2

FIT Strategies (v2)
WTP = $16M per QALY

	Cost (M$)	QALYs	ICER (M$)
NoScreening	263.4	0.0	—
FIT5570q3	276.5	25.5	0.51
FIT5070q3	279.6	30.9	0.58
FIT4570q3	290.3	37.0	—
FIT5070q2	291.1	37.5	1.74
FIT4570q2	304.8	42.8	2.59
FIT4575q2	316.7	45.0	5.41
FIT4570q1	367.0	51.6	7.64
FIT4575q1	383.5	52.8	13.48
FIT4580q1	396.4	53.5	19.08
FIT4585q1	405.2	53.7	40.83



Costs v3

FIT Strategies (v3)
WTP = $16M per QALY

	Cost (M$)	QALYs	ICER (M$)
NoScreening	268.5	0.0	—
FIT5570q3	339.8	25.5	2.79
FIT5070q3	356.1	30.9	3.02
FIT4570q3	387.4	37.0	5.14
FIT5070q2	394.6	37.5	—
FIT4570q2	430.1	42.8	7.33
FIT4575q2	455.6	45.0	11.58
FIT4570q1	563.4	51.6	16.38
FIT4575q1	595.8	52.8	26.56
FIT4580q1	620.2	53.5	36.07
FIT4585q1	636.3	53.7	74.06

